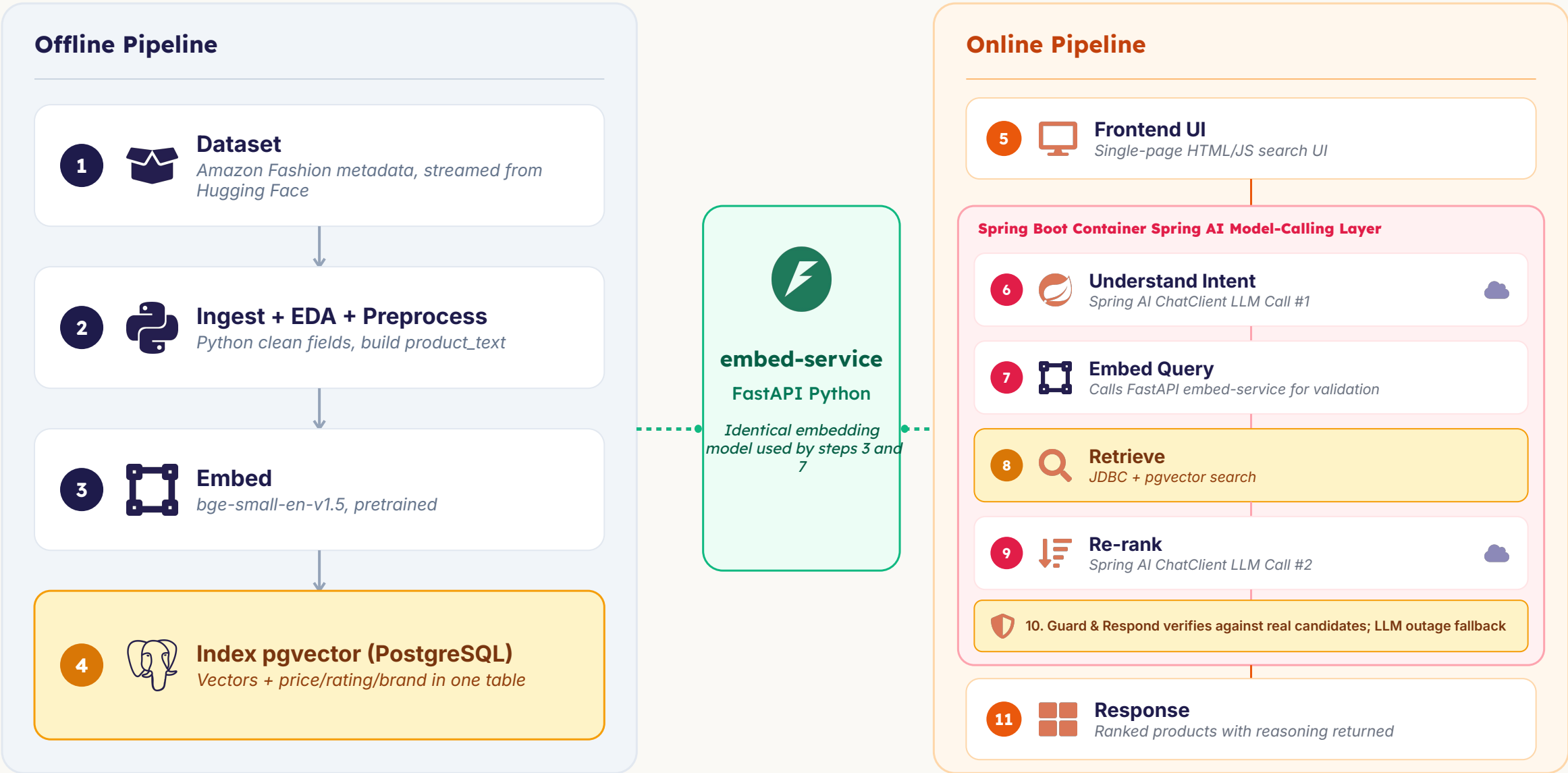


Semantic Recommendation Microservice

Architecture offline training, online serving. Spring AI is the only place a model is called.



Database Alignment: The online Retrieve step (8) reads the exact index and PostgreSQL instance that the offline pipeline (4) populates.

OFFLINE FLOW DESCRIPTION

- 1. Dataset:** Real Amazon Fashion product metadata streamed live from Hugging Face no synthetic or placeholder data.
- 2. Ingest + EDA + Preprocess:** Clean inconsistent fields (price as string; details as JSON), strip HTML, and build one consolidated product_text per item.
- 3. Embed:** Encode every product with a pretrained sentence-transformer into a 384-dimensional vector. Fine-tuning is a documented next phase.
- 4. Index:** Store vectors alongside price, rating, and brand in a single PostgreSQL/pgvector table for hybrid search filtering.

ONLINE FLOW DESCRIPTION

- 5. Frontend UI:** A single-page HTML/JS UI submits queries to the REST API without complex builds or frameworks.
- 6. Understand Intent:** Spring AI ChatClient rewrites the query and extracts category/price constraints only if explicitly stated.
- 7. Embed Query:** Java calls the external FastAPI embed-service over HTTP to guarantee identical model execution as step 3.
- 8. Retrieve:** Performs a vector similarity search on the pgvector table, pre-filtered by the extracted user constraints.
- 9. Re-rank:** Second LLM call evaluates and orders top candidates, outputting a precise grounded rationale.
- 10. Guard & Respond:** Verifies results against candidate sets; falls back gracefully to vector search in case of LLM outages.
- 11. Response:** Returns final list with structured reasoning directly to the client UI.